

PORTLAND HILLSBORO, OR, USA

WMO: 726986

Lat: 45.541N

Lon: 122.949W

Elev: 204

StdP: 14.59

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94261

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.1	26.6	11.6	10.0	30.0	17.9	13.6	33.3	23.3	44.2	20.2	41.9	3.2	310	0.505

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	92.4	67.6	88.3	66.5	84.1	65.0	69.7	87.6	67.9	85.3	66.1	82.1	7.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	87.5	75.9	61.3	81.6	73.0	59.7	77.1	71.2	33.6	87.9	32.1	85.0	30.8	82.3	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.4	16.3	13.7	DB	16.8	99.4	5.2	3.6	13.1	102.0	10.0	104.1	7.1	106.2	3.4	108.8
			WB	15.9	74.4	5.1	3.7	12.2	77.1	9.2	79.2	6.3	81.3	2.6	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	40.7	42.5	46.3	50.3	56.7	61.3	67.4	67.4	62.0	52.6	44.9	39.8
	DBStd	11.29	6.21	5.53	5.12	5.54	6.07	5.82	5.56	5.13	5.64	5.34	6.20	6.29
	HDD50	1245	293	215	138	60	10	0	0	0	1	34	175	319
	HDD65	4768	754	631	579	443	266	141	34	31	122	384	603	780
	CDD50	2241	4	4	24	68	218	341	540	539	362	115	22	4
	CDD65	290	0	0	0	1	9	31	109	106	33	1	0	0
	CDH74	4746	0	0	1	50	258	567	1642	1526	674	28	0	0
	CDH80	1918	0	0	0	10	71	218	718	657	241	3	0	0
(14) Wind	WSAvg	4.8	4.6	5.1	5.5	5.3	4.9	5.0	5.2	4.7	4.2	4.0	4.6	4.8
(15) Precipitation	PrecAvg	43.5	6.9	4.4	4.4	3.2	2.5	1.6	0.4	0.7	1.4	3.7	6.6	7.5
	PrecMax	67.6	14.1	10.8	7.8	6.7	5.2	3.8	1.7	2.7	6.3	8.8	15.6	15.7
	PrecMin	32.4	2.6	0.2	1.5	1.2	0.0	0.2	0.0	0.0	0.0	0.6	2.2	2.2
	PrecStd	8.3	2.6	2.3	1.7	1.5	1.4	0.8	0.4	0.6	1.2	1.9	3.1	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.5	61.2	70.4	80.8	86.5	93.5	98.8	97.1	92.1	78.3	63.5	57.5
		MCWB	54.1	51.0	55.3	62.2	65.0	68.2	70.0	69.9	65.9	60.0	56.9	54.3
	2%	DB	54.5	57.1	64.3	72.8	81.3	86.4	92.6	91.7	86.6	72.0	60.0	54.2
		MCWB	50.8	50.0	52.8	58.2	62.2	66.0	68.2	67.0	63.8	57.7	55.4	51.9
	5%	DB	52.0	54.3	59.5	66.4	75.4	80.7	88.0	87.6	81.8	67.2	57.0	51.6
		MCWB	48.9	48.4	50.4	54.5	60.1	63.8	67.2	66.6	63.2	55.7	53.0	49.4
	10%	DB	49.8	51.8	55.9	61.5	70.1	75.0	83.0	82.4	76.9	63.1	54.5	48.5
		MCWB	47.2	47.1	49.5	52.0	57.9	61.2	65.5	65.2	61.5	55.4	51.0	46.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.3	55.7	57.3	64.0	67.3	70.8	73.0	72.7	69.0	64.4	59.6	55.1
		MCDB	57.5	57.8	66.5	77.7	84.0	88.7	92.3	89.7	86.1	72.5	61.9	56.6
	2%	WB	52.0	51.9	54.4	59.0	63.9	67.1	69.8	69.2	66.2	60.4	56.6	52.1
		MCDB	53.5	55.1	60.4	70.4	77.2	84.1	87.8	87.2	81.8	66.6	59.0	53.7
	5%	WB	49.6	49.7	52.3	55.6	61.4	64.3	67.9	67.4	64.3	58.3	54.1	49.5
		MCDB	51.3	53.1	57.8	64.3	73.3	78.4	85.9	84.2	78.1	64.0	56.1	51.1
	10%	WB	47.5	47.4	50.5	53.1	58.9	61.9	65.9	65.6	62.4	56.5	51.8	46.6
		MCDB	49.3	50.7	55.5	60.1	68.9	74.0	82.2	81.4	74.0	61.6	53.8	48.0
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	14.9	17.1	19.3	21.2	22.5	27.5	27.8	26.2	19.7	13.9	10.9
		MCDBR	13.1	19.5	24.8	30.9	32.9	33.6	36.3	36.7	36.2	29.2	16.5	11.8
	5% WB	MCWBR	9.6	12.6	14.3	17.2	17.0	16.4	15.6	15.9	16.8	16.3	11.2	9.5
		MCWBR	10.9	14.5	20.3	26.9	28.8	30.8	33.3	33.1	31.4	21.0	12.6	10.9
(40) Clear-Sky Solar Irradiance	taub		0.320	0.316	0.330	0.358	0.372	0.361	0.349	0.354	0.342	0.331	0.323	0.325
		taud	2.531	2.553	2.520	2.404	2.383	2.426	2.487	2.485	2.508	2.546	2.530	2.477
	Ebn at Noon		252	277	287	286	284	287	289	284	279	267	248	234
		Edh at Noon	21	24	29	35	37	36	33	32	29	24	21	20
(44) All-Sky Solar Radiation	RadAvg		368	635	923	1345	1713	1893	2158	1842	1377	796	430	300
	RadStd		60	89	122	126	166	177	151	123	121	100	59	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	-0.86	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+41

Nomenclature: See separate page